

Exploring Review Delivery Contexts in VR Shopping: A Study Design

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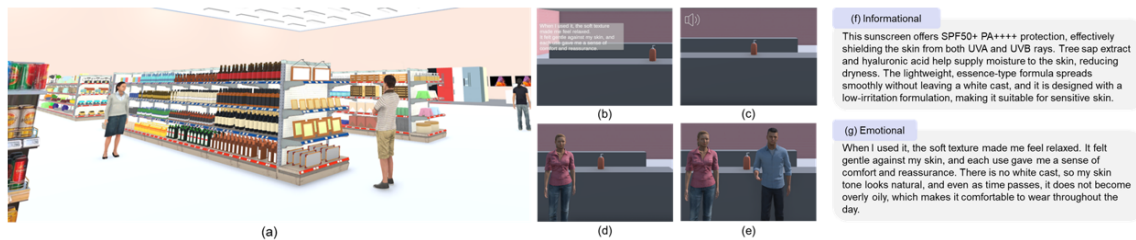


Figure 1: (a) Virtual shopping environment used in the study. The study employs a 4 (review delivery context: (b) text, (c) audio, (d) single agent, (e) multiple agents) $\times$ 2 (review message type: (f) informational, (g) emotional) factorial design. (f) and (g) present example informational and emotional review messages, respectively.

ABSTRACT

As VR shopping environments increasingly support social presence and embodied interaction, consumer reviews may serve not only as sources of information but also as socially situated cues. This study explores review delivery in VR shopping from a socially embedded perspective and proposes an experimental design to examine how delivery context and message characteristics may influence user judgment and conformity. By framing review delivery context as a novel design dimension in VR shopping environments, this work is expected to offer implications for marketing, VR commerce, and research on persuasion mediated by virtual agents.

Index Terms: VR Shopping, Review Delivery Context, Social Conformity, Virtual Agents, Decision-Making.

1 INTRODUCTION

In online shopping environments, the inability to directly experience products compels consumers to rely on the evaluations and opinions of others. In this context, consumer reviews function not merely as supplementary information but as vital social cues that mitigate uncertainty and legitimize purchase decisions [12]. While online reviews have traditionally been delivered in text-based formats, their expansion into audio and video has demonstrated that the same information can elicit varying levels of trust and purchase intention depending on how the information is presented [2].

Recently, virtual reality (VR)-based shopping environments have extended these experiences into a new dimension. By integrating multi-sensory stimuli with spatial interaction, VR shopping offers a higher degree of immersion and presence than conventional interfaces, enabling users to explore products within a simulated physical space [7]. Consequently, VR environments serve not only as platforms for information access but also as social arenas where consumer-agent interactions can emerge.

Critically, in VR, reviews can be delivered through the speech and behaviors of embodied virtual agents—a feature that distinguishes VR from traditional online settings. Within an immersive space, an agent's utterances may be perceived as originating from a social entity, allowing reviews to operate as social signals rather

than neutral data. While prior studies have established that social cues influence consumer conformity [11], this line of inquiry has largely been confined to non-immersive or 2D interfaces.

Moreover, existing research has primarily focused on how content characteristics—such as informational versus affective reviews—shape judgment [12, 10], leaving the social context of delivery relatively underexplored. Specifically, it remains unclear how users respond to the same message when it is transitioned from text to audio, or delivered by single versus multiple virtual agents.

To address this gap, this study conceptualizes review delivery in VR shopping not merely as a choice of modality, but as a social delivery context. By structuring delivery mechanisms—ranging from text to multiple-agents configurations—along a continuous spectrum, we aim to examine how these contexts shape user conformity and purchase decisions. Furthermore, by incorporating both informational and emotional messages, we examine the interaction between message type and delivery context to provide a comprehensive understanding of consumer decision-making in VR. Based on these considerations, we formed following research questions:

RQ1. How does review delivery context, ranging from non-social to agent-mediated configurations, shape user conformity and purchase-related decision-making in VR shopping?

RQ2. How does the effectiveness of review message types (informational vs. emotional) vary across different delivery contexts?

RQ3. To what extent does the interaction between delivery context and message type influence consumer decision-making in immersive VR environments?

2 MATERIALS

To address research questions, we use review delivery context and review message type as core experimental design factors.

Review Delivery Context Review delivery context is defined as the social and interactional configuration through which review information is conveyed to the user. Rather than treating review presentation as a simple modality choice, delivery context captures varying degrees of social embeddedness. Prior research has shown that social cues and the presence of others can significantly influence judgment and conformity [4, 3]. In immersive VR, embodied virtual agents may be perceived as social actors, thereby amplifying such effects through increased social presence [13, 9]. Building on these findings, this study proposes multiple review delivery contexts along a spectrum ranging from non-social text-based presentation to agent-mediated and group-based configurations.

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Review Message Type Review messages are designed to be generated in two forms: informational and emotional. Informational reviews emphasize objective product attributes and factual descriptions, whereas emotional reviews focus on subjective experiences and affective reactions. This distinction is well established in prior electronic word-of-mouth (eWOM) and advertising research, which suggests that informational and emotional messages influence consumer judgment through different cognitive and affective mechanisms [12, 8, 5]. In the present study, semantic meaning, length, and evaluative valence are controlled across message types to isolate the effects of message type and delivery context.

3 METHOD: PLANNED STUDY

The present study adopts a 4×2 factorial design to investigate how review delivery context and review message type influence decision-making in VR shopping contexts (See Fig. 1). **Review delivery context** is manipulated at four levels: text-based reviews, audio-based reviews without an embodied agent, single-agent delivery, and multiple-agents delivery—representing increasing degrees of social embeddedness in review presentation. **Review message type** is manipulated at two levels, consisting of informational and emotional reviews, which reflect distinct cognitive and affective modes of evaluation commonly discussed in prior eWOM research.

To systematically examine the effects of review delivery context and message type on decision-making, the study employs a controlled product evaluation task in which participants will be exposed to reviews for products from the same category under different conditions. Participants experienced all experimental conditions sequentially, enabling within-subject comparisons across delivery contexts and message types. The order of the eight experimental conditions was counterbalanced across participants using a balanced Latin square design [1].

Across all conditions, the same 3D product model was used, and only the review delivery context and message type varied. For the agent-based conditions, embodied virtual agents will be used to verbally present reviews. The agents are built using a publicly available Microsoft-provided rigged avatar library [6] designed for VR research and applications. To enhance realism, the agents will differ in appearance and animation, and agent gender will be randomized across conditions.

Based on this design, we proposed the following hypotheses:

- **H1.** Agent-based reviews are expected to elicit higher perceived social presence than text-based reviews.
- **H2.** Agent-based reviews will exert a stronger influence on purchase intention than non-agent-based reviews.
- **H3.** Emotional reviews will foster higher perceived trust when delivered in agent-based contexts than in non-agent-based contexts.

Procedure The study will be conducted in a Unity-based VR environment simulating a virtual retail store. Participants will navigate the environment through a guided exploration path and will be exposed to product reviews for a fixed duration (30 seconds) in each condition. During this phase, participants will rate their purchase intention using a 7-point Likert scale. After experiencing all experimental conditions, they will be given additional time to freely explore the virtual store and then make a final purchase decision by selecting one product.

Measures Participants will report basic demographic information prior to the experiment. The primary dependent variables—perceived social presence and purchase intention—will be measured using questionnaires adapted from prior VR and consumer behavior research. Several secondary measures, including cognitive workload (NASA-TLX), affect (Mehrabian affect scale),

simulator sickness (SSQ), and perceived presence (IPQ), will be collected to assess participant experience and experimental validity. In addition, the system will log decision time during the selection phase as a behavioral measure.

4 CONCLUSION

This paper presents a study design to investigate the effects of review delivery context and review message type on consumer decision-making in VR shopping environments. While the planned study primarily relies on self-report measures, future studies could further enhance validity by incorporating eye-tracking data to capture users' visual attention during decision-making.

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